Number Codes (Set B)

Year 4 Pt 2 · **Lesson Worksheet**

Worked Examples

Example 1: A number lock uses this code: 2 = H, 5 = O, 8 = T. What is the code for 258?

Answer: HOT — 2 = H, 5 = O, 8 = T, so 258 → H, O, T.

Example 2: A code is formed by doubling a number and then adding 1. What is the code for 7?

Answer: 15 — 7 × 2 = 14, then 14 + 1 = 15.

Example 3: In a code, every digit has 2 added to it, wrapping back round to 0 after 9 (so 9 becomes 1). What is the code for 793?

Answer: 915 — 7 → 9, 9 → 11 → 1, 3 → 5, giving 915.

Purple — Accessible (15 questions)

1. A suitcase lock uses this code: 3 = K, 6 = P, 8 = T, 9 = R. What is the code for the number 368?
2. A walkie-talkie uses this code: 1 = M, 4 = S, 5 = L, 7 = D. The code DLS stands for which number?
3. A safe uses this code: 1 = L, 3 = O, 5 = C, 7 = K. What is the code for the number 357? A) OKC B) OCK C) COK D) KOC E) CKO
4. A treasure map uses this code: 2 = G, 3 = O, 5 = N, 9 = V. What is the code for the number 293?
5. A robot's control panel uses this code: 0 = Z, 4 = Q, 6 = X, 8 = H. The code XQH stands for which number?
6. A bike lock uses this code: 2 = M, 4 = I, 6 = X, 8 = T. The code MIT stands for which number? A) 284 B) 428 C) 824 D) 248 E) 482
7. A pirate's chest uses this code: 1 = A, 2 = B, 7 = Y, 9 = Z. What is the code for the number 192?
8. A spy radio uses this code: 3 = F, 4 = J, 5 = K, 6 = W. The code JWK stands for which number?
9. Using the code 0 = G, 1 = T, 2 = O, 3 = P, 4 = N, 5 = W, 6 = F, 7 = S, 8 = L, 9 = M, select ALL the numbers below whose code would begin with the letter 'T'. A) 184 B) 253 C) 102 D) 719 E) 136
10. A garage door uses this code: 0 = B, 3 = E, 5 = D, 9 = G. What is the code for the number 903? A) GBE B) BGE C) EBG D) GEB E) BEG
11. A vending machine uses this code: 0 = C, 2 = E, 5 = I, 7 = U. What is the code for the number 207?
12. A garden gate keypad uses this code: 1 = T, 3 = R, 6 = E, 8 = N. The code NET stands for which number?
13. A phone passcode uses this code: 1 = P, 2 = A, 4 = S, 6 = T. The code SAP stands for which number? A) 124 B) 214 C) 421 D) 412 E) 241
14. A castle drawbridge uses this code: 2 = D, 4 = R, 5 = A, 9 = W. What is the code for the number 954?
15. A school locker uses this code: 1 = S, 2 = U, 3 = N. The code SUN stands for which number?

Green — Standard (15 questions)

1. A code is formed by doubling a number and then adding 5. What is the code for 9?

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2. A code is formed by multiplying a number by 3 and then subtracting 2. The code for a certain number is 22. What was the original number?

3. A code is formed by doubling a number and then subtracting 3. What is the code for 8? A) 11 B) 12 C) 13 D) 14 E) 16

4. A code is formed by adding 4 to a number and then doubling the result. What is the code for 6?

5. A code is formed by halving a number and then adding 7. The code for a number is 12. What was the original number?

6. A code is formed by multiplying a number by 4 and then adding 2. The code for a number is 30. What was the original number? A) 7 B) 8 C) 6 D) 9 E) 5

7. In a code, 3 → 11 and 5 → 19. The same rule is used for every number. What is the code for 7?

8. In a code, 4 → 14 and 9 → 24. The same rule applies to every number. If the code for a certain number is 20, what is that number?

9. A code is formed by multiplying a number by 3 and then adding 1. Select ALL the numbers below whose code would be greater than 20. A) 5 B) 6 C) 7 D) 8 E) 9

10. In a code, 5 → 17 and 8 → 26. Using the same rule, what is the code for 12? A) 34 B) 35 C) 36 D) 37 E) 38

11. A code is formed by subtracting 3 from a number and then multiplying the result by 3. What is the code for 10?

12. A code is formed by multiplying a number by 5 and then adding 1. The code for a number is 31. What was the original number?

13. A code is formed by subtracting 2 from a number and then multiplying by 4. What is the code for 9? A) 24 B) 28 C) 32 D) 20 E) 26

14. In a code, 2 → 8 and 6 → 20. What is the code for 10, using the same rule?

15. In a code, 3 → 8 and 7 → 16. If the code for a number is 12, what is that number, using the same rule?

Red — Challenge (15 questions)

1. In a code, every digit of a number is replaced by the digit that is 3 more than it (if the digit goes above 9, wrap back round to 0 — so 9 becomes 2, for example). What is the code for 486?

2. Using the same rule (add 3 to each digit, wrapping round after 9), the code for a number is 052. What was the original number?

3. In a code, every digit has 5 added to it, wrapping round after 9. What is the code for 385? A) 835 B) 380 C) 083 D) 830 E) 038

4. A code is made by reversing the digits of a number and then adding 1 to the result. What is the code for 123?

5. Using the same rule (reverse the digits, then add 1), the code for a number is 456. What was the original number?

6. A code is made by reversing the digits of a number and then subtracting 2. The code for a number is 140. What was the original number? A) 241 B) 214 C) 412 D) 421 E) 124

7. In a code, each digit is replaced by a letter (4 = M, 7 = I, 1 = N), and then the resulting letters are reversed. What is the code for 471?

8. Using the same type of code (substitute each digit using 6 = T, 2 = A, 9 = P, then reverse the letters), the code for a number is PAT. What was the original number?

9. In a code, every digit has 6 added to it, wrapping round after 9 (so a digit of 4 becomes 0). Select ALL the numbers below whose code would contain the digit 0. A) 142 B) 253 C) 489 D) 617 E) 954

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10. In a code, 573 is written as KMV, and 735 is written as MVK. Using the same code, how would 357 be written? A) KVM B) VKM C) MKV D) KMV E) VMK

11. In a code, 341 is written as DGT, and 134 is written as TDG. Using the same code, how would 413 be written?

12. In a code, 258 is written as WLR, and 825 is written as RWL. Using the same code, which number is written as LWR?

13. In a code, every digit has 4 added to it, wrapping round after 9. The code for a number is 207. What was the original number? A) 386 B) 638 C) 683 D) 368 E) 863

14. In a code, every digit has 7 added to it, wrapping back round after 9 (so 9 becomes 6: 9 + 7 = 16, 16 − 10 = 6). What is the code for 362?

15. Using the same rule (add 7 to each digit, wrapping round after 9), the code for a number is 146. What was the original number?

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